

ZEBRA RFID AI PROJECT NOTE

Zebra RFID AI Application & SDK Sample

Android RFID Application & Gemini AI Developer Assistant

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Project architecture, setup requirements, build instructions, and Gemini AI code assistant notes for the Zebra RFID Android application.

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An Android application demonstrating production-ready integration with the **Zebra RFID API3 SDK** (`com.zebra.rfid.api3`) alongside **Gemini 3.8 AI Assistant** built with **Jetpack Compose Material 3**.

Application Screenshots

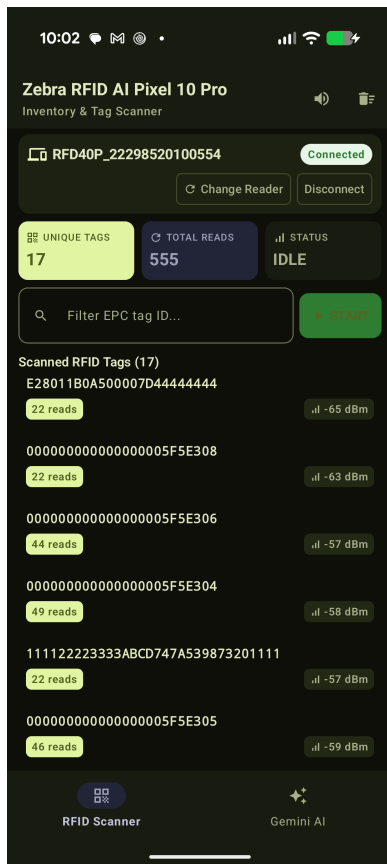


Figure 1: RFID Inventory Scanner

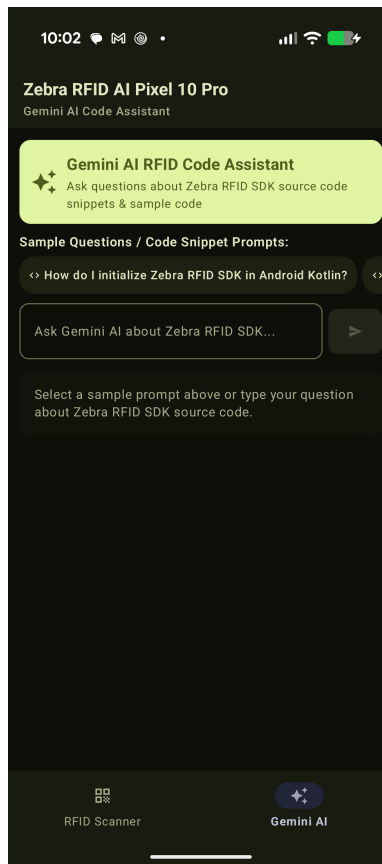


Figure 2: Gemini AI Code Assistant

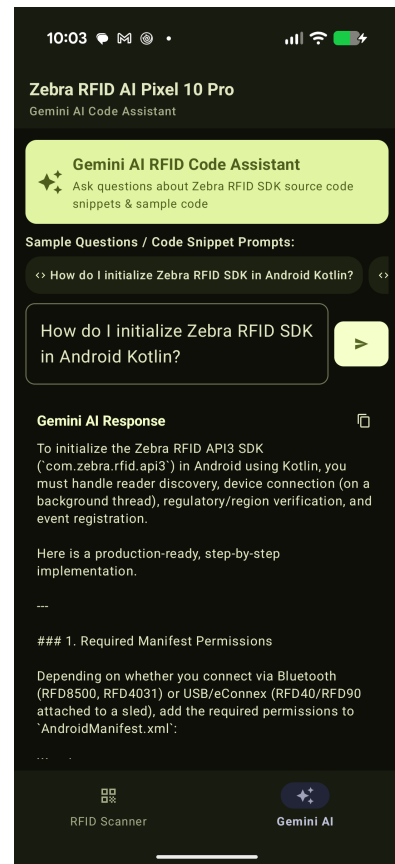


Figure 3: Gemini AI Code Response

Features

- **Zebra RFID SDK Integration (`rfdapi3lib-2.0.5.292.aar`)**
 - Thread-serialized background SDK execution (`ExecutorService`) preventing UI thread blocking.
 - Automatic transport scanning across available reader interfaces.
 - Reader discovery, auto-connect, and region configuration (**USA** regulatory default).
- **Lifecycle-Aware Suspend & Resume**
 - **Suspend:** Automatically disconnects reader and disposes `Readers` on `ON_PAUSE` / `ON_STOP`.
 - **Resume:** Re-initializes SDK, discovers readers, and reconnects seamlessly on `ON_RESUME` / `ON_START`.
- **Real-Time RFID Tag Inventory**

- Displays unique tag count, total read count, live scan status badge, and EPC tag items.
- Calculates per-tag read counts and peak RSSI values (dBm).
- Physical handheld trigger support (starts inventory on press, stops inventory on release).
- Real-time search/filter by EPC tag ID.
- **Audible Tone Feedback**
 - **ToneGenerator** audio beeps for reader connection, disconnection, and new tag discovery.
- **Gemini 3.8 AI Assistant**
 - **Code Assistant:** Ask questions or request code snippet explanations for Zebra RFID SDK APIs (**RFIDReader**, **Readers**, **RfidEventsListener**, **TagData**).
 - **Inventory Analysis:** Analyzes active inventory tag lists directly with Gemini AI.
 - **Response Caching:** In-memory and persistent **SharedPreferences** caching for instant AI responses on repeated queries.

Project Architecture

```

app/src/main/java/com/zebra/rfid/ai/
├── MainActivity.kt           # Entry point, Bluetooth/Location permissions &
└── LifecycleObserver
├── MainAppScreen.kt         # TopAppBar & Bottom NavigationBar (Scanner vs Gemini AI)
├── RfidScreen.kt            # Material 3 UI for RFID Connection, Stats, & Tag List
├── RfidViewModel.kt         # ViewModel & StateFlow for Tag Aggregation & UI State
├── RfidHandler.kt           # Thread-serialized Zebra RFID SDK Manager (Init, Connect,
└── Suspend, Inventory)
├── RfidUtility.kt           # Helper methods for Reader Discovery, Regions, & Events
├── ToneFeedback.kt          # Audio feedback generator
├── RfidAiScreen.kt          # Gemini AI UI with Preset Prompts & Code Explanations
└── RfidAiViewModel.kt      # Firebase Gemini 3.8 AI ViewModel with Local Caching

```

Getting Started

Prerequisites

- **Android Studio** 2026.1+ with JDK 19+.
- **Android Device:** Zebra TC701 / RFD40 / RFD8500 or Android device running API 35+.
- **Permissions:** Bluetooth (**BLUETOOTH_SCAN**, **BLUETOOTH_CONNECT**) and Location.

Building & Running

1. Clone the repository:

```

git clone https://github.com/GelatoCookie/Rfid-AI.git
cd Rfid-AI

```

2. Build the debug APK:

```
./gradlew app:assembleDebug
```

3. Install and run on device:

```
./gradlew app:installDebug
```

License

Licensed under the Apache License, Version 2.0.